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LifeTrail: Benchmarking Multimodal Memory, Personalization, and Proactive Assistance over Long-Horizon Real-World Life Traces

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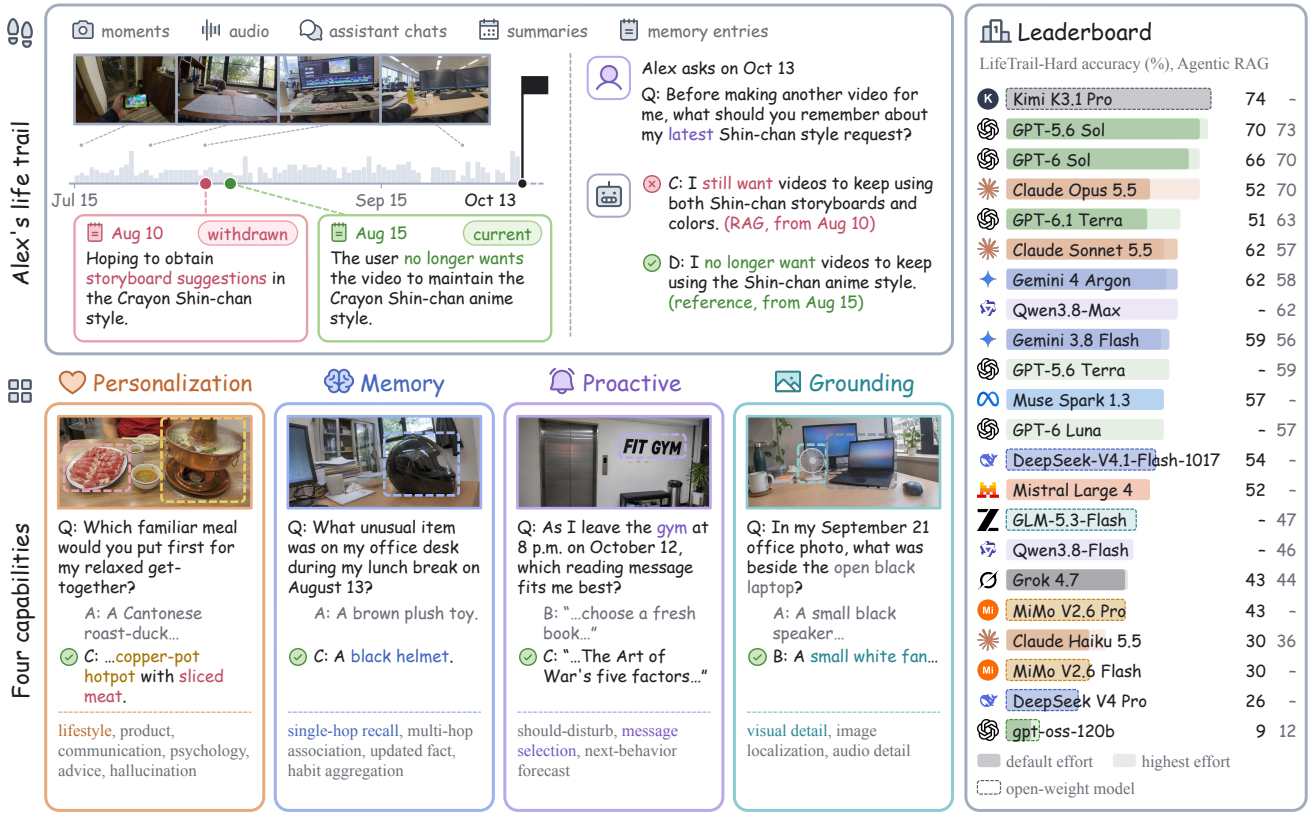


Figure 1. **Overview of LifeTrail.** LifeTrail asks questions over a user's multimodal life records in LifeAtlas to evaluate how well personal assistants understand the user's long-term life. Assistants must track evolving preferences across large volumes of image-text records and keep their memory up to date. Top left: In a three-month life trail, a request made on Aug 10 is withdrawn on Aug 15, yet when asked about the user's latest request on Oct 13, RAG-based answer models still select the withdrawn one. Bottom left: The four capabilities that LifeTrail evaluates (personalization, memory, proactive assistance, and multimodal grounding), with example questions. Right: Accuracy of 22 models on LifeTrail-Hard with Agentic RAG, at the default reasoning effort of each API (deep bars) and at the highest effort (pale bars). Dashed outlines mark open-weight models.

Abstract

Personal AI assistants must connect a user's everyday experience with their preferences, evolving circumstances, and immediate needs. Existing benchmarks evaluate personalized assistants on chat histories, synthetic profiles, or short recordings rather than on months of real-world experience. We collect LifeAtlas, a dataset of more than 180k multimodal life records, including photos, audio transcripts, locations, and assistant chats, from more than 150 users over up to three months. On top of it, we build LifeTrail, a benchmark of over 1,300 multiple-choice questions spanning 16 tasks in four capabilities: personalization, memory, proactive assistance, and multimodal grounding. An agentic pipeline automates question construction, and each released question is reviewed by the user whose records it uses. We evaluate memory systems and retrieval methods with three answer models. Memory systems that consoli-

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date the history perform best, yet every setting remains at least 15 points below gold evidence, which makes retrieving the relevant records the main bottleneck of personal agent systems. Agentic RAG improves factual recall over reranked multimodal RAG but lowers accuracy on personalization and updated facts. Proactive inference remains difficult even with gold evidence, and higher reasoning effort brings the largest gains on these questions. We further benchmark 22 frontier models on the challenging LifeTrail-Hard under varying reasoning effort and tool budgets. The strongest model reaches 74% accuracy, and higher reasoning effort helps more than a larger tool budget. The dataset and benchmark will be released upon publication.

1. Introduction

Personal AI assistants are moving from answering isolated requests to acting for a user over weeks and months. Agents built on frontier models now search large personal archives, call tools, and plan multi-step actions [45, 93], and a growing number of memory systems give them persistent state across sessions [14, 61, 63, 87]. Useful assistance in this setting requires recalling events from the user's past, tracking how their preferences change, and deciding when an intervention would help [10, 47, 51]. Such systems appear far faster than benchmarks that test them on real users, and comparing them requires longitudinal records of everyday experience that capture how a user's circumstances and preferences change over time.

Wearable lifelogging cameras capture everyday experience through first-person images and audio. These data can be organized into records of conversations, locations, and recurring activities [22, 23, 27, 91], which reveal how a user's preferences and intentions evolve, including cases where newer information contradicts an earlier request. In Fig. 1 (top left), a user requests cartoon-style storyboards on August 10 and withdraws the request five days later. Two months later, the user asks what to keep in mind before making the next video. With RAG-based retrieval, all three answer models select the withdrawn request. A record that was correct when it was made thus leads to a wrong personalized answer once a later change is overlooked.

Existing benchmarks evaluate different aspects of personal memory and reasoning. Dialogue-memory benchmarks emphasize recall and temporal reasoning over text interaction histories [17, 54, 82], personalization benchmarks often rely on synthetic user profiles [34, 35, 59], and egocentric benchmarks focus on visual understanding of first-person recordings [9, 12, 55, 91]. Most of these benchmarks rely on chat histories, synthetic profiles, or short recordings rather than months of real everyday experience, and they evaluate each capability in isolation without testing how memory, personalization, and proactive assistance interact on the same history.

To close this gap, we collect **LifeAtlas**, a dataset of multimodal life records from more than 150 users who wore a lifelogging camera for up to three months. On top of LifeAtlas, we build **LifeTrail**, a benchmark of over 1,300 multiple-choice questions spanning 16 tasks in four capabilities: *personalization*, *memory*, *proactive assistance*, and *multimodal grounding* (Fig. 3 and Table 1). Every question fixes an anchor time and allows only earlier records. Writing such questions by hand does not scale to months of records from many users, so we automate construction with an agentic pipeline (Fig. 2). It samples task-specific evidence, writes first-person questions grounded in the records, and verifies each candidate with independent agentic checks. Each question that passes is then reviewed by the user whose records it uses.

Our contributions are:

- **Dataset.** LifeAtlas, over 180k real-world multimodal life records from more than 150 users over up to three months (Sec. 3.1), which supports research on memory systems, egocentric understanding, and personal agents.
- **Benchmark.** LifeTrail, 16 tasks across four capabilities that share the same histories and temporal access rule, and the challenging LifeTrail-Hard (Secs. 3.2 and 4). Its agentic data engine needs no manual question writing and applies to other collections of personal records.
- **Evaluation.** A systematic study of memory systems, retrieval methods, and 22 frontier models under controlled reasoning effort and tool budgets (Sec. 5). It shows that locating the relevant records is the main bottleneck, that agentic retrieval gains factual recall at the cost of personalization, and that proactive inference requires reasoning beyond the retrieved evidence.

2. Related Work

Long-Term Memory. Long-term memory benchmarks test whether an assistant can retain and update information over extended interaction histories, mostly in text dialogue [86, 101]. LoCoMo [54] evaluates recall and temporal reasoning over multi-session conversations, and PerLTQA [17] evaluates them over mixed episodic and semantic memories. LongMemEval [82] adds knowledge updates and abstention, MemoryAgentBench [31] evaluates selective forgetting under incremental input, and LifeDialBench [105] restricts each query to the preceding history. Multimodal benchmarks and systems ground memory in captured photos and videos [43], images from wearable devices [36], streaming video and audio [50], conflicting records [7], and changes in visual state [26]. ATM-Bench [57], the closest to our setting in data and scale, poses 1,038 human-written questions that resolve personal references over about four years of a person's photos, videos, and emails. LifeAtlas in-

Table 1. **Comparison of LifeTrail with related benchmarks.** The columns follow the memory abilities of Guo *et al*. [25]: cross-modal evidence (CE), long-range evidence (LE), state changes (SC), conflicting memories (CF), unanswerable queries (UQ), and memory-grounded action (MA), plus personalization (👤) and proactive assistance (⏰), added by us. ✓, ○, and ✗ mark direct, partial, and no evaluation. Ground truth: answer (A), evidence records (E), timestamps (T), actions or states (S), rubric (R).

Benchmark	Source	Span	Ground Truth	CE	LE	SC	CF	UQ	MA	👤	⏰
Image and conversational memory											
LoCoMo [54]	Hybrid	months	A+E	○	✓	○	✗	✓	✗	✗	✗
Memory-QA [36]	Real	months	A+E	○	✓	✗	✗	✗	✗	✗	✗
ATM-Bench [57]	Real	years	A+E	✓	✓	✓	✓	✓	✗	✓	✗
Mem-Gallery [7]	Synth.	Unknown	A+E	✓	✓	✓	✓	✓	✗	○	✗
MemEye [26]	Hybrid	Unknown	A+E	✓	✓	✓	✗	✗	✗	✗	✗
PersonaMem-v3† [35]	Hybrid	30 days	A+R	✗	✓	✓	○	✓	✓	✓	✓
Egocentric video											
EgoLifeQA [91]	Real	1 week	A+T	✓	✓	○	✗	✗	✗	✓	✗
S-EMBER [78]	Real	Unknown	A+E+T	○	✓	○	✗	✗	✗	✗	✗
EgoMonth [12]	Real	20–120 days	A+T	○	✓	✓	✗	✗	✗	✗	✗
Personal agents											
KnowU-Bench [11]	Hybrid	Unknown	S	✓	✓	✓	✗	✗	✓	✓	✓
MEMORA-Bench [98]	Real	Unknown	A+S	○	✓	○	✗	✓	✓	✓	✗
LifeTrail (ours)	Real	3 months	A+E+T	○	✓	✓	○	✓	○	✓	✓

stead covers up to three months from more than 150 users, and LifeTrail evaluates preference inference and proactive assistance together with recall on the same histories.

Memory systems differ in how they build memory and retrieve relevant information for a query. MemGPT [61] and MemoryOS [38] manage memory hierarchies, Zep [66] maintains a temporal knowledge graph, Mem0 [14] keeps compact, incrementally updated facts, A-MEM [87] links notes, MemFit [63] indexes verbatim records with segment summaries, and ViLoMem [8] accumulates multimodal semantic memory. Retrieval methods range from text and multimodal embeddings [19, 37, 42, 44, 65] with LLM reranking [73] to graph- and tree-based indexes [18, 28, 41, 68], learned retrieval decisions [3], structured memory navigation [88], and agentic search with tools [45, 93], which has also been applied to video event memories [46, 49, 94–96]. These designs trade compactness against fidelity: LongMemEval reports that fact compression improves multi-session reasoning but lowers overall accuracy on dialogue histories. We evaluate representatives of each design on real lifelogs against a gold-evidence setting, and Table 1 places LifeTrail in the memory-ability framework of Guo *et al*. [25].

Lifelogging and Egocentric Understanding. Lifelogging studies the capture and retrieval of real-world experience recorded by wearable devices [22, 27, 74], in which egocentric data play an increasingly important role. Egocentric datasets have expanded from kitchen and skilled activities [15, 24] to daily life recorded with multi-sensor wearables [52, 53], and benchmarks have lengthened the horizon and broadened the tasks. Ego4D [23] introduced episodic-memory queries, later extended to question answering [6, 33] and to interactive retrieval with user feedback [72]. EgoSchema [55], HourVideo [9], and general long-video benchmarks [20, 107] extend question answering to long-form footage, where agents search captions or frames [77, 80, 99]. Work on individual histories includes single-life visual learning [29], wearer-centric question answering [85], streaming retrieval [78], and multi-day memory [81]. EgoLife [91] records six participants sharing one week and evaluates recall, habits, and recommendations, while EgoMonth [12] evaluates spatiotemporal memory over 20–120 days. Other benchmarks target in-the-wild assistance [89], real-time task help [21], audio-visual understanding [70], and first-person web tasks [97].

Personalized LLMs and MLLMs. Personalization methods adapt model parameters, augment inputs with user context, or align outputs with preferences [59, 83, 102]. Adaptation methods learn user-specific visual concepts from images [1, 2, 58, 62] and videos [71], reuse the model's own concept representations [69], or share concepts across understanding, generation, and editing [106]. Among augmentation methods, RAP [30] retrieves concepts from an editable database, and PersonaVLM [59] couples multimodal memory with personality-aware generation. Alignment methods such as Personalized Soups [32] merge models trained for separate preference dimensions. On the evaluation side, text benchmarks condition on personas or user profiles [40, 67, 100] or on preferences stated in conversation [104]; PersonaMem [34] tests responses to evolving synthetic profiles, and LifeVal [92] evaluates the completion of personalized consumer tasks. LifeTrail provides no profile or concept set: models must infer preferences and habits from the recorded history.

Proactive Assistance. A proactive assistant decides whether and when to act and what message to deliver [16], and receptivity to such interruptions has long been studied for mobile notifications [56]. Proactive Agent [51] studies task initiation from observed user activity. Streaming video models decide when to respond [10, 64, 79] and are evaluated on online understanding [48, 60] and response timing [4]. Benchmarks and methods for egocentric task guidance [5, 76], next-step planning [13], and action anticipation [103] focus on a single recorded activity. ProactEgoMem [47] tests the retention of transient information and object states in egocentric streams, and PersonaMem-v3 [35] evaluates cross-platform recommendation, including when to refrain. KnowU-Bench [11] and ProactiveMobile [39] ask mobile agents to anticipate needs from behavioral histories, and MEMORA [98] pairs egocentric recall with planning. LifeTrail evaluates interruption decisions, message selection, and next-behavior forecasting on real users' lifelogs across diverse daily scenarios, on the same histories as its memory and personalization tasks.

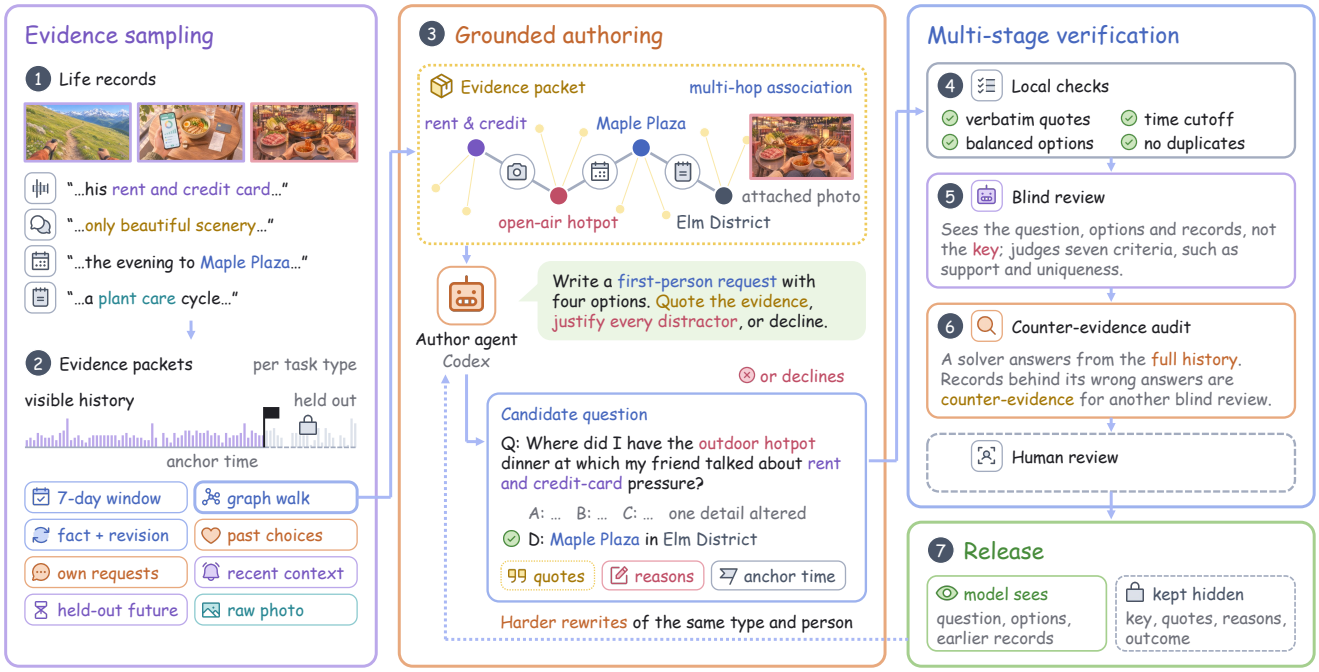


Figure 2. **Overview of the LifeTrail construction pipeline.** Each user's life records are split at an anchor time into a visible history and a held-out future, from which task-specific samplers draw evidence packets (1-2). An author agent then writes a first-person question grounded in the packet (3). Candidates pass multi-stage verification, including local checks, blind review, counter-evidence audit and human review (4-6). At release, models see only the question, options and earlier records (7).

3. Dataset and Benchmark

3.1. The LifeAtlas Dataset

LifeAtlas contains the histories of more than 150 users who wore a lifelogging camera between mid July and mid October 2026. Its basic unit is the *moment*, a continuous activity segment with photos, an audio transcript, a location, and timestamps. Histories also contain assistant chats, daily and weekly summaries, and memory entries distilled from them. In total, LifeAtlas holds over 180k records, including over 95k moments that contain over 250k photos. All users gave informed consent to the research use of their records and are identified only by pseudonyms. Before release, all photos and transcripts are anonymized in compliance with the GDPR (supplementary Sec. A).

3.2. Task Definition

Each user u has a history $\mathcal{H}_u = \{r_i\}_{i=1}^{N_u}$, where a record $r_i = (\tau_i, x_i)$ pairs its end time τ_i with multimodal content x_i, such as photos, an audio transcript, a location, or text. A question $q = (u, t_q, x_q, \mathcal{O}_q)$ consists of an anchor time t_q, a first-person request x_q, and an option set $\mathcal{O}_q$ with four options, or two for most should-disturb questions. A model f observes only the visible history $\mathcal{H}_u^{<t_q} = \{r_i \in \mathcal{H}_u : \tau_i < t_q\}$ and selects

$$\hat{o}_q = f\big(x_q, \mathcal{O}_q, \mathcal{H}_u^{<t_q}\big) \in \mathcal{O}_q, \tag{1}$$

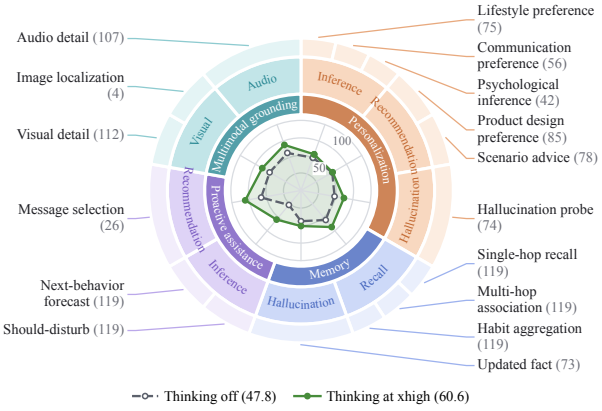


Figure 3. **Task taxonomy of LifeTrail.** The rings show the four capabilities (inner), the nine task groups of Table 2 (middle), and the 16 tasks with their question counts (outer). The radar in the center gives the accuracy of Qwen3.8-Flash in the Agentic RAG evaluation setting on each task group, with thinking off and at the xhigh reasoning effort.

which is correct if it equals the reference option o_q^*. The 16 tasks share this interface and differ in how the supporting records E_q determine o_q^* (Fig. 3).

Personalization asks for the option most consistent with the user's preferences and traits θ_u, $o_q^* =$

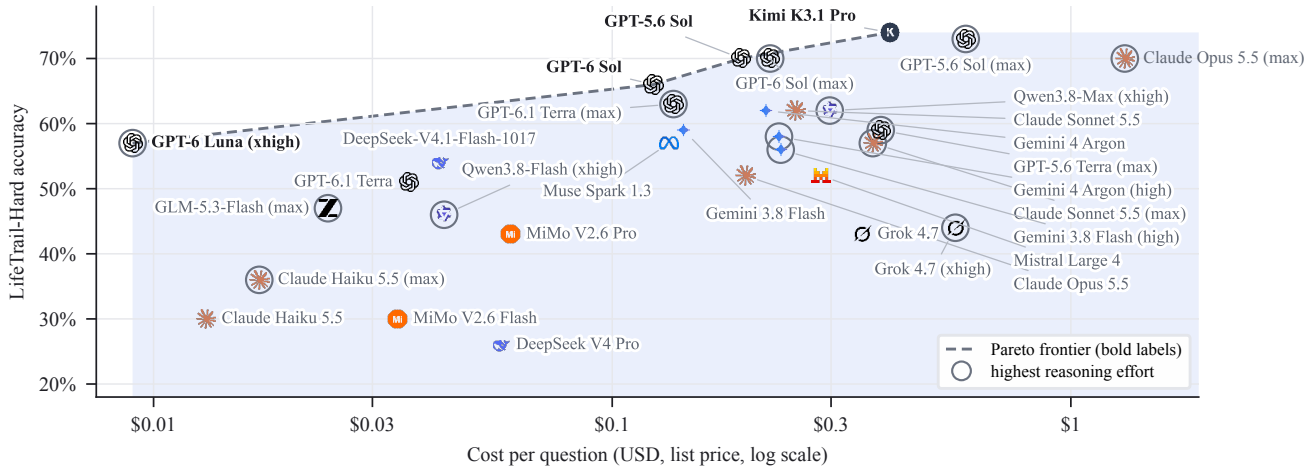


Figure 4. **Accuracy versus average task cost on LifeTrail-Hard.** All models are evaluated with Agentic RAG and a tool-call limit of 15. We omit gpt-oss-120b, which reaches at most 12%. Plain logos mark the default effort of each model and ringed logos its highest effort. The dashed line is the Pareto frontier (bold labels).

$\arg\max_{o \in \mathcal{O}_q} s(o; \theta_u)$, where s measures consistency and θ_u is inferred from many records in E_q rather than read from one record. A hallucination probe embeds a preference that no record in $\mathcal{H}_u^{<t_q}$ supports, and o_q^* states that the history does not support it.

Memory asks for a fact stated in the history, $o_q^* = g(E_q)$ with $E_q \subseteq \mathcal{H}_u^{<t_q}$, where E_q is one record for single-hop recall, a chain of records for multi-hop association, and all instances in a window for habit aggregation. For an updated fact, $E_q = \{r_a, r_b\}$ with $\tau_a < \tau_b < t_q$ and r_b revising r_a, so o_q^* follows r_b and a distractor follows r_a.

Proactive Assistance asks for a decision at t_q from the recent context $\mathcal{C}_q = \{r_i : t_q - \delta \le \tau_i < t_q\}$ and the earlier history. Should-disturb decides whether to interrupt at t_q, and message selection chooses the message that fits $\mathcal{C}_q$ and θ_u. Next-behavior forecast asks for the activity $a(r^+)$ of a held-out record r^+ with $\tau^+ > t_q$, which f never observes.

Multimodal Grounding asks for a detail determined by a photo whose text description is withheld (visual detail), by the record whose photo matches a query photo (image localization), or by an audio transcript (audio detail). Labels for memory, grounding, and forecasting questions follow directly from the records, whereas labels for personalization and the two proactive judgments are inferred from the user's behavior and confirmed in the human review of Sec. 4.2.

3.3. Benchmark Characteristics

Table 1 compares LifeTrail with related benchmarks on six memory abilities [25]. Cross-day association tests long-range evidence (LE); revised facts test state changes and conflicting memories (SC, CF); hallucination probes test unanswerable queries (UQ). Proactive tasks extend evaluation toward memory-grounded action (MA), and grounding tasks require visual or transcribed-audio evidence (CE). Because all tasks share the same histories and temporal access rule, capabilities can be compared for the same user. For 533 of the 1,327 questions, the earliest supporting record precedes the question by more than 60 days. Multiple-choice answers make scoring deterministic and comparable across models without a language-model judge.

4. Agentic Data Engine

Unchecked generation produces questions that are unsupported, ambiguous, or answerable without the user's history. To avoid these failures at scale, we design an agentic data engine that grounds every question in sampled evidence and filters every candidate with independent agentic checks before human review (Fig. 2).

4.1. Grounded Construction by Evidence Sampling

Task-specific samplers draw evidence packets around an anchor time, such as a chain of linked records for multi-hop association or a statement and its revision for updated facts; proactive packets follow ContextAgent [90]. A Codex agent with GPT-6.1 Sol writes first-person questions from each packet, with the correct option and verbatim evidence quotes. Every distractor is plausible for the same user and comes with a rationale that cites the evidence against it.

4.2. Multi-Stage Verification

Local Checks enforce the question format, verbatim quotes, and the anchor-time rule.

Blind Review. A fresh GPT-6.1 Sol session receives the question, options, and permitted records, without the reference answer. It must select the author's option and confirm that the answer is supported, unique, and not leaked.

Counter-Evidence Audit. Qwen3.8-Flash answers each surviving question from the user's full visible history.

Table 2. **Main results on LifeTrail.** Accuracy (%) ranges from no evidence (lower bound) to gold evidence (upper bound). Columns are the task groups of Fig. 3, where memory Hallu. holds the updated facts. Qwen and DeepSeek are Qwen3.8-Flash and DeepSeek-V4.1-Flash without thinking, and Luna is GPT-6 Luna with medium thinking. Highlighted cells are the best in each column for each answer model between the two bounds, and ▲/▼ mark Agentic RAG at least 8 points above or below reranked Multimodal RAG.

	Method	Answerer	Personalization			Memory		Proactive		Grounding		Overall ↑
			Infer.	Rec.	Hallu.	Recall	Hallu.	Infer.	Rec.	Visual	Audio	
Lower	No evidence	Qwen	50.3	46.0	71.6	34.7	45.2	17.6	53.8	33.6	43.0	38.7
		DeepSeek	49.7	46.6	52.7	33.9	52.1	20.6	53.8	38.8	36.4	38.2
		Luna	53.2	48.5	58.1	37.8	53.4	41.2	61.5	34.5	41.1	44.2
Memory	A-MEM [87]	Qwen	**66.6**	**62.6**	59.2	**58.4**	**65.8**	19.2	**68.5**	44.7	**78.7**	**54.0**
		DeepSeek	60.9	55.7	55.1	51.1	62.8	29.1	68.0	49.1	**73.7**	51.9
		Luna	**78.2**	**71.8**	**86.9**	**67.1**	69.2	53.6	89.9	**71.3**	**86.0**	**70.2**
	Mem0 [14]	Qwen	64.8	57.6	60.9	55.9	60.4	16.6	62.2	44.1	76.8	51.5
		DeepSeek	67.8	**68.9**	**60.9**	**58.7**	64.0	26.7	**82.4**	46.1	69.0	**56.0**
		Luna	75.0	70.4	82.4	66.6	69.8	**58.6**	**91.0**	51.5	80.9	68.1
	MemFit [63]	Qwen	52.0	57.1	52.7	43.4	53.4	17.6	57.7	44.8	63.6	44.7
		DeepSeek	59.0	56.4	56.8	47.3	61.6	29.4	69.2	55.2	64.5	50.6
		Luna	65.3	62.6	85.1	54.6	65.8	56.7	88.5	60.3	72.9	62.3
Retrieval	Naive RAG	Qwen	65.9	60.7	59.5	51.0	61.6	16.4	57.7	48.3	64.5	50.0
		DeepSeek	69.9	61.3	60.8	49.9	61.6	30.3	69.2	53.4	57.9	53.0
		Luna	75.1	65.0	74.3	61.1	72.6	52.5	76.9	69.8	73.8	65.3
	Multimodal RAG	Qwen	64.2	55.8	63.5	44.0	58.9	18.1	61.5	46.6	43.9	45.9
		DeepSeek	66.5	58.3	60.8	42.3	63.0	30.3	65.4	45.7	44.9	48.4
		Luna	68.8	62.6	75.7	49.6	60.3	49.6	69.2	47.4	52.3	56.1
	+ Rerank	Qwen	65.3	62.0	**64.9**	44.0	64.4	18.5	65.4	39.7	57.9	47.9
		DeepSeek	**71.1**	61.3	59.5	43.4	**67.1**	29.0	73.1	44.8	49.5	50.0
		Luna	73.4	65.0	82.4	47.9	**74.0**	51.7	76.9	50.0	57.0	58.9
Agent	Agentic RAG	Qwen	49.7 ▼	51.5 ▼	48.6 ▼	54.9 ▲	43.8 ▼	**26.9** ▲	57.7	**51.7** ▲	57.0	47.8
		DeepSeek	57.8 ▼	52.1 ▼	58.1	55.5 ▲	46.6 ▼	**31.1**	65.4	**63.8** ▲	64.5 ▲	52.3
		Luna	52.0 ▼	43.6 ▼	60.8 ▼	59.7 ▲	38.4 ▼	34.9 ▼	61.5 ▼	61.2 ▲	65.4 ▲	51.8
Upper	Gold evidence	Qwen	87.9	92.0	82.4	77.6	91.8	27.7	88.5	80.2	94.4	74.6
		DeepSeek	87.3	87.7	85.1	74.2	93.2	33.6	88.5	82.8	92.5	74.5
		Luna	90.2	93.9	91.9	93.3	91.8	54.2	96.2	91.4	93.5	85.7

Records cited in its wrong answers are added as counter-evidence to a second blind review, which rejects the question if another option becomes defensible.

Human Review. Each released question-answer pair is reviewed by the user whose records it uses, who checks the answer against the cited records.

4.3. Question Selection and LifeTrail-Hard

From over 5k questions accepted by the automated stages and their harder rewrites, the final selection balances user and task coverage and favors questions that GPT-6.1 Sol answers incorrectly in a difficulty screening, which yields 1,327 questions from 154 users. Because GPT-6.1 Sol both writes the questions and screens them for difficulty, the benchmark is biased against it, and we exclude it from evaluation. For the frontier-model study, **LifeTrail-Hard** contains 100 questions from 57 users that cover all 16 tasks and are selected by their error rate across screening settings. With gold evidence, Qwen3.8-Flash answers all 100 questions correctly, so their difficulty lies in finding and combining the evidence rather than in the labels.

5. Experiments

We systematically compare memory and retrieval settings on the 1,327 questions of LifeTrail and benchmark mainstream frontier models on the 100 questions of LifeTrail-Hard under varying reasoning effort and tool budgets to understand how test-time scaling affects them.

5.1. Experimental Setup

Evidence Settings. *No evidence* supplies only the question and options. A-MEM [87], Mem0 [14], and MemFit [63] build their own memories from the visible records.

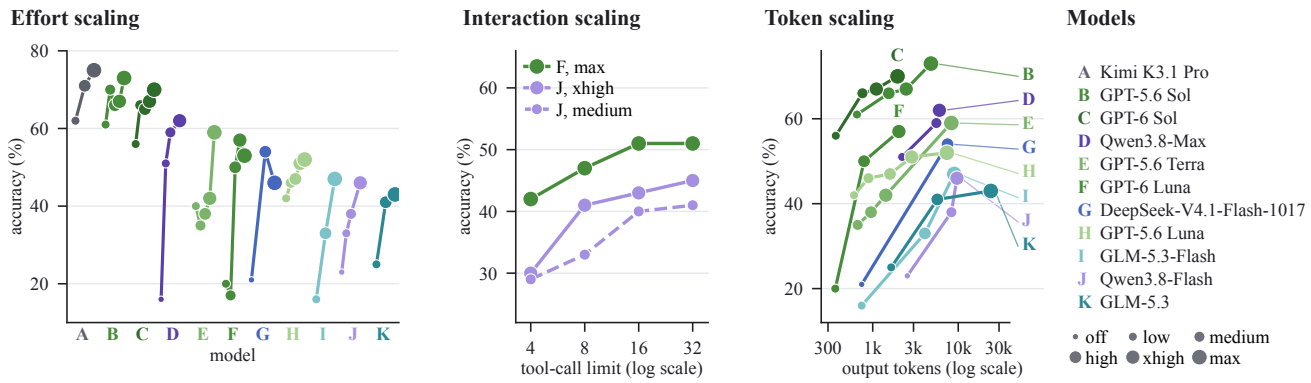


Figure 5. **Test-time scaling on LifeTrail-Hard.** All models use Agentic RAG. Letters identify the models listed on the right, and larger dots mean higher reasoning effort. Left: Each curve shows the accuracy of one model from its lowest to its highest effort. Middle: Accuracy as the tool-call limit grows from 4 to 32. Right: Accuracy against output tokens per question.

Naive RAG returns ten records ranked by text and photo-caption embeddings, with their photos, and *Multimodal RAG* uses fused text-image embeddings [44], optionally reranked. *Agentic RAG* searches moments and opens selected records with up to 15 tool calls. *Gold evidence* supplies the reference records.

Models and Scoring. The full 1,327-question evaluation uses Qwen3.8-Flash and DeepSeek-V4.1-Flash without thinking, and GPT-6 Luna with medium thinking. On LifeTrail-Hard, we compare 22 models from 11 providers at their default and highest reasoning efforts (Fig. 1, right) and report their cost, following Collaborative Intelligence [84]. Empty or unparseable responses count as wrong. Because most should-disturb questions have two options, the chance level of proactive inference is 36.1%, compared with 25% for the other task groups.

5.2. Main Results

Evidence Access. For each answer model, the highest overall accuracy comes from A-MEM or Mem0, which consolidate the history into linked notes or compact facts before answering. Yet every memory and retrieval setting stays at least 15 points below gold evidence (Table 2). The same answer models answer most questions correctly once the reference records are supplied, so the main difficulty of LifeTrail lies in locating the relevant records within months of history rather than in answering from them. Among the RAG settings, Naive RAG passes about twice as much context as reranked Multimodal RAG and reaches the highest overall accuracy for all three answer models, which shows that broad coverage of the history matters more than finer ranking of a few records.

Capability-Specific Evidence. Agentic RAG improves memory recall and visual grounding over reranked Multimodal RAG for all three answer models, but it lowers accuracy on personalization and on updated facts, the memory hallucination group (▲/▼ in Table 2). These capabilities rest on different evidence. A recall or grounding question is answered by one or a few records that match its keywords, whereas a preference is expressed across many records, and an updated fact requires the later of two conflicting records (Sec. 3.2). Agentic search suits questions whose answer lies in a few matching records and falls short on questions that require aggregating many records or identifying the most recent one. Conversely, A-MEM and Mem0, which consolidate many records into notes or facts, lead most personalization columns for every answer model.

Reasoning Beyond Evidence. Proactive inference, which covers should-disturb and next-behavior forecast, is the only task group on which gold evidence leaves accuracy low: even with gold evidence, Qwen3.8-Flash and DeepSeek-V4.1-Flash stay below its 36.1% chance level. These questions ask what the user will do or accept next, which no record states, so the model must infer it from the user's current situation and habits. Reasoning effort supports this inference. On the full benchmark, thinking at xhigh raises the proactive accuracy of Qwen3.8-Flash with Agentic RAG by 26.9 points, the largest gain of any capability, and improves every task group (Fig. 3).

5.3. Additional Analysis

Answer Changes with Evidence. Following the memory lifecycle analysis of Guo *et al.* [25], we classify each question by comparing the answers of the same model with no evidence, in a given setting, and with gold evidence. Compared with reranked Multimodal RAG, Agentic RAG overturns two to three times as many personalization answers that were correct without evidence, while the share of personalization errors that gold evidence would recover stays nearly unchanged. Its personalization loss thus comes from how the retrieved records are weighed rather than from missing evidence. On proactive questions, errors that gold evidence would recover stay below 14% in every setting, and most errors persist with gold evidence, so better re-

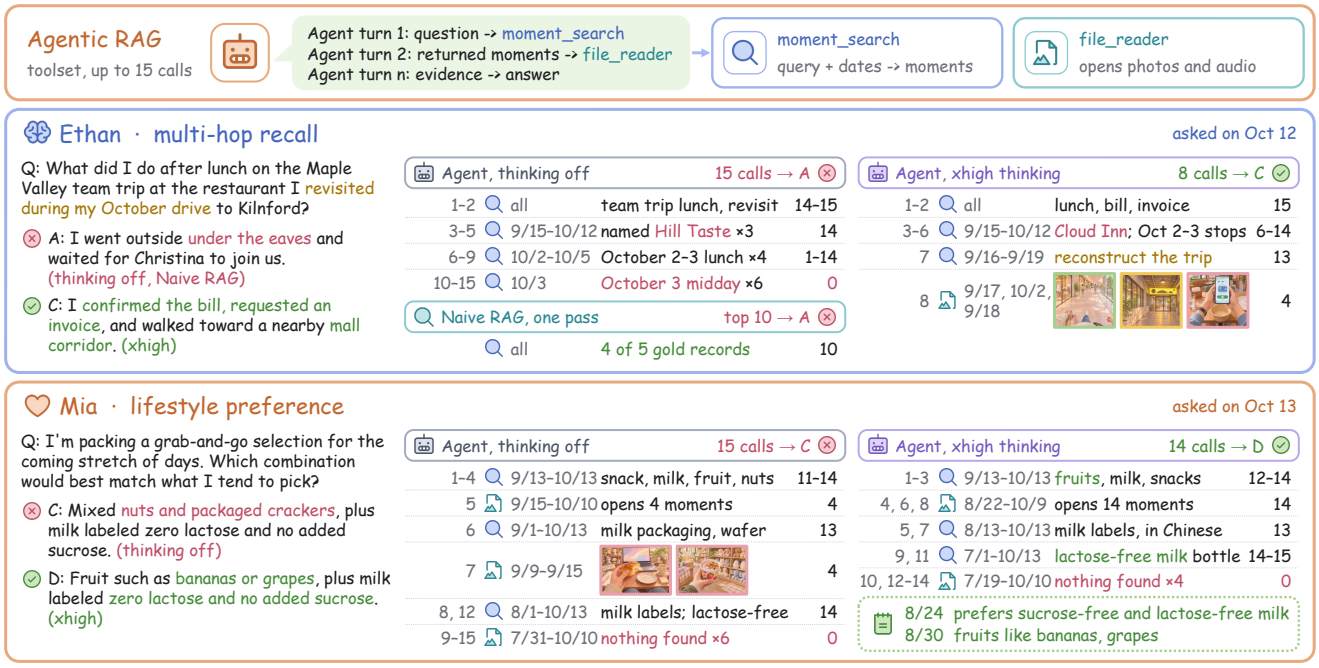


Figure 6. **Visual examples of Agentic RAG on LifeTrail.** We show traces of the Qwen3.8-Flash agent with thinking off and at xhigh. Top: Multi-hop recall. Naive RAG retrieves most of the gold records yet answers incorrectly. Without thinking, the agent spends nine of its 15 calls on an unrecorded period, whereas at xhigh it reconstructs the trip from two connecting moments. Bottom: Lifestyle preference. The preference is stated only in memory entries outside the search index, and only the xhigh agent opens enough moments to recover it.

trieval alone does not close this gap.

Effort and Tool Budget. The strongest model on LifeTrail-Hard, the open-weight Kimi K3.1 Pro, answers 74% of the questions (Fig. 1, right). As in MCP-Persona [75], accuracy does not follow cost (Fig. 4): Claude Opus 5.5 at max effort costs about seven times as much as GPT-5.6 Sol at medium effort for the same accuracy. We further vary reasoning effort, the tool-call limit, and output tokens (Fig. 5). Higher effort raises accuracy for most models, while raising the tool-call limit from 16 to 32 changes accuracy by at most two points (Fig. 5, middle), so the agent is limited by reasoning rather than by its search budget. With thinking at xhigh, Qwen3.8-Flash halves its tool calls on the full benchmark, and its traces show that thinking makes the search more selective rather than longer (Fig. 6). Higher effort also turns unanswered questions into correct answers: from low to xhigh effort, GPT-6 Luna answers 37 more questions correctly, while its wrong choices stay almost constant (34 and 33).

6. Conclusion

We present LifeAtlas, a dataset of real-world multimodal life records from more than 150 users over up to three months, and LifeTrail, a benchmark of 1,327 multiple-choice questions on personalization, memory, proactive assistance, and multimodal grounding, together with the more challenging LifeTrail-Hard. An agentic pipeline grounds each question in the user's records before an anchor time, and each released question is reviewed by the user whose records it uses. Experiments with memory systems, retrieval methods, and mainstream frontier models under controlled reasoning effort and tool budgets show that retrieving the relevant records from months of history is the main bottleneck, that agentic retrieval improves factual recall but lowers personalization accuracy, and that proactive assistance requires reasoning beyond the gold evidence.

As personal assistants move from answering isolated requests to acting for a user over months, these findings outline what they require. Their memory must follow how the user's context changes, so that a withdrawn request or a changed habit supersedes the earlier record, and it must aggregate the many weak cues through which preferences are expressed. Their reasoning must interpret the retrieved evidence about the user correctly and exercise restraint, neither attributing unsupported preferences to the user nor interrupting when an interruption would not help. Because such memory holds months of a person's life, it must also remain under the user's control, so that users can inspect, correct, and delete what the assistant remembers. By releasing LifeAtlas and LifeTrail under strict anonymization, we provide the community with longitudinal records of real lives and an effective evaluation platform for future research on personal intelligence, toward assistants that remember their users accurately and act in their interest.

LLM Usage Statement

Large language models are part of the method and the object of study in this work. In data construction, GPT-6.1 Sol in a Codex agent writes candidate questions from sampled evidence and performs the blind reviews, Qwen3.8-Flash performs the counter-evidence audit, and language models translate the records into English and audit each translation. Every released question is reviewed by the user whose records it uses, and GPT-6.1 Sol is excluded from evaluation. The photos shown in our figures are generated images that reproduce the recorded scenes, so no photo of a real user or bystander appears in the paper. We used coding assistants (Claude Code and Codex) to assist in implementing the pipeline and the figure scripts. We also used language-model assistants to edit the text and to check its consistency with the tables and figures. The authors reviewed all generated code and text, verified every number against the experiment logs, and take full responsibility for the content of the paper.

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